

Proofs

Theorem 1.1. (The Division Theorem for $\mathbb{Z}$): $a, b \in \mathbb{Z}$ with $b > 0$ $\exists!$ $q, r \in \mathbb{Z}$ with $0 \leq r < b$ s.t. $a = bq + r$.

Proof. (Existence): Let $R = \{a - bm : m \in \mathbb{Z} \text{ and } a - bm \geq 0\}$, then as: if $a \geq 0$, $a - b \cdot 0 = a \geq 0$ so $a \in R$; if $a < 0$, $a - b \cdot a = a(1 - b) \geq 0$ so $a - b \cdot a \in R$, $R \neq \emptyset$. Thus, take $r = \min(R)$ and so

$0 \leq r = a - bm \in R$ for some $m \in \mathbb{Z}$, choose $q = m \in \mathbb{Z}$. So by construction $a = bq + r$. Note, $r < b$ as were such not true then $r > a - b(q + 1) = r - b \in R$ but this contradicts $r = \min(R)$, so we have $0 \leq r < b$. (Uniqueness): Suppose $q', r', s, r' \in \mathbb{Z}$ with $0 \leq r', r' < b$ and

$a = bq + r = bq' + r'$, also WLOG take $r' \leq r$. So with $\frac{r-r'}{b} \in (-1, 1)$, &

$\frac{r-r'}{b} = \frac{a-bq-bq'}{b} = q' - q \in \mathbb{Z}$, we have $\frac{r-r'}{b} = 0$ i.e. $r = r'$. So

$q = \frac{a-r}{b} = \frac{a-r'}{b} = q'$.

Euclidean Algorithm:

Let $a, b \in \mathbb{Z}^+$, apply Th^m 1.1, to give:

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ &\vdots \end{aligned}$$

$r_{n-1} = r_nq_{n+1}$, Here, $\gcd(a, b) = r_n$. To prove this relation we show:

$G(r_i, r_{i+1}) = G(r_{i+1}, r_{i+2})$, where $G(r, s) = \{n \in \mathbb{Z} : n|a \text{ \& } n|b\}$.

Reverse Euclid:

Given a completed algorithm we can obtain $n, m \in \mathbb{Z}$ s.t.

$na + mb = \gcd(a, b)$ i.e. taking the start of a Euclidean algorithm:

$$\begin{aligned} a &= q_1b + r_1 & \Rightarrow r_1 &= a - q_1b \\ b &= q_2r_1 + r_2 & \Rightarrow r_2 &= b - q_2r_1 \\ &= b - q_2(a - q_1b) \\ &= -q_2a + (1 + q_1q_2)b \\ r_1 &= q_3r_2 + r_3 & \Rightarrow r_3 &= r_1 - q_3r_2 \\ &= (a - q_1b) \\ &\quad - q_3(-q_2a + (1 + q_1q_2)b) \\ &= (1 + q_2q_3)a \\ &\quad - (q_1 + q_3 + q_1q_2q_3)b \\ &\vdots \end{aligned}$$

Continue substituting until only a, b remain:

$$r_n = \gcd(a, b) = na - mb, \quad n, m \in \mathbb{Z}.$$

Theorem 1.9. Every integer $n \geq 2$ can be written as a product of primes.

Proof. (By strong induction):

(Base case, $n = 2$): Trivially true, $n = 2$ a product of primes.

(Inductive Hypothesis): Suppose for $n \in \mathbb{Z}, 3, \dots, k$

$n = p_{n1}p_{n2} \dots p_{n,m_n}$.

Then for $n = k + 1$, if $k + 1$ is prime we are done. If not,

$k + 1 = ab$ where

$a, b \in \{2, 3, \dots, k\}$, so by the hypothesis a and b are products of primes; hence $k + 1$ is too.

So $\forall n \in \mathbb{N}$ with $n \geq 2$, we are done.

Theorem 1.10. (Euclid's property of primes) $\forall p \in \text{Primes } \forall a, b \in \mathbb{Z}$. $p|ab \Rightarrow p|a \text{ or } p|b$.

Proof. (Case $p|a$): We are done. (Case $p \nmid a$): Then using the Euclidean algorithm we obtain:

$1 = na - mp$ with $n, m \in \mathbb{Z}$. So

$p|b \Leftrightarrow p|(na - mp) \Leftrightarrow$

$p|n(ab) - mbp \Leftrightarrow p|ab$.

the chain of implications uses a proposition in the notes which states:

$p|a \text{ \& } p|b \Rightarrow p|a + b \text{ \& } p|ab$.

Theorem 1.12. (Fundamental Theorem of Arithmetic) $\forall n \in \mathbb{N}$ with $n \geq 2$, $\exists!$ $p_1, p_2, \dots, p_r \in \text{Primes}$ up to ordering, s.t.: $n = p_1p_2 \dots p_r$.

Proof. The existence is from Th^m 1.9. For uniqueness, suppose

$p_1p_2 \dots p_r = q_1q_2 \dots q_m$, then cancel common primes, suppose the result is not 1 i.e. we are left with

$p_1p_2 \dots p_n = q_1q_2 \dots q_m$ with $p_i \neq q_j, \forall i \in \{1, \dots, n\} \text{ \& } \forall j \in \{1, \dots, m\}$, after reindexing. Then $p_1|q_1q_2 \dots q_m$ so

$\exists j \in \{1, \dots, m\} p_1|q_j$, but q_j is prime so $p_1 = q_j$.

Lemma 1.16. $\forall a, b \in \mathbb{R}$: $0a = a0 = 0$; $a(-b) = (-a)b = -(ab)$; $\mathcal{G}(-a)(-b) = ab$

Proof. $0a + 0 = 0a = (0 + 0)a = 0a + 0a$ so $0a = 0$.

$ab + a(-b) = a(b - b) = a0 = 0 = (a + a)b = ab + (-a)b$ so by the Groups & Geo^m we have $-a$ is the unique element s.t.

$ab - ab = 0$,

$a(-b) = -(ab) = (-a)b$.

$(-a)(-b) = -(-a(-b)) = -(-(ab)) = ab$.

Lemma 1.18. (Subring Test) Let R be a ring $\mathcal{G} S \subseteq R$. Then S is a subring of R iff the following hold, $\forall r, s \in S$:

1. $1_R \in S$;

2. $r + s, rs \in S$, $\mathcal{G}$

Proof. By 1 & 3 we have $-1_R \in S$ so $0_R = 1_R - 1_R \in S$ by 2. So $(S, +)$ is a group by the subgroup test with $(R, +)$. Commutativity of $+$, associativity of $+$ & $\times$, &

distributivity of $\times$ over $+$ are all inherited from R . Note, as they share operations they share the identities of the operations, so as,

$0_R \neq 1_R$ in R we have

$0_S = 0_R \neq 1_R = 1_S$ in S , so

(R1)-(R4) are satisfied. $\square$

Lemma 2.2. Suppose $\text{Char}(R) = n > 0$, then: (i) $n \cdot r = 0, \forall r \in R$;

(ii) if $m \in \mathbb{Z}^+$ then $m \cdot 1 = 0$ iff $n|m$.

Proof. (i):

$n \cdot r = r + r + \dots + r$

$= r(1 + 1 + \dots + 1) = r0 = 0$

(ii): ($\Rightarrow$): Let $m \cdot 1 = 0$ with $m \in \mathbb{Z}^+$, applying the division

Th^m with $m = nq + r$ where

$q, r \in \mathbb{Z}$ & $0 \leq r < n$ so

$0 = m \cdot 1 = (nq + r) \cdot 1 =$

$q(n \cdot 1) + r \cdot 1 = q \cdot 0 + r \cdot 1 = r \cdot 1$,

so $r \cdot 1 = 0$ as otherwise $n > r \in \mathbb{Z}^+$ & $r \cdot 1 = 0$, a $\mathcal{G}$ of n 's defⁿ.

($\Leftarrow$): Let $n|m$ i.e. $\exists q \in \mathbb{Z}$ s.t. $m = qn$, so $m \cdot 1 = qn \cdot 1 =$

$q(n \cdot 1) = q \cdot 0 = 0$.

Lemma 2.5. If S is a subring of R $\mathcal{G}$ S is a domain then S is a domain.

Proof. Let $r, s \in S$ with $rs = 0$, thus $rs = 0$ in R , hence $r = 0$ or $s = 0$.

Proposition 2.7. Suppose R is a domain. Then $R[X]$ is a domain.

Proof. We use the contrapositive of the definition of a domain [i.e.: R a domain iff $\forall 0 \neq r, s \in R, rs \neq 0$].

Let $0 \neq f, g \in R[X]$, taking the leading coefficients of f & g as $a \neq 0$ & $b \neq 0$, respectively, the leading coefficient of fg as $ab \neq 0$, hence $fg \neq 0$.

Note: This immediately (by induction on n) yields

$R[X_1, \dots, X_n]$ is a domain if R is a domain.

Lemma 2.11. If R is a ring $\mathcal{G} r \in R$ has both a left $\mathcal{G}$ right inverse then these are equal $\mathcal{G}$ unique.

Proof. (Equal): Choose, $rs = 1$ tr , then:

$s = 1s = trs = t1 = t$.

(Unique right inverse): Let

$rs = 1 = rs'$ then $s = 1s = trs =$

$t1 = trs' = 1s' = s'$, where t is chosen s.t. $tr = 1$.

(Unique left inverse): Let

$tr = 1 = t'r$ then $t = t1 = trs =$

$1s = t'sr = t'1 = t'$, where s is chosen s.t. $rs = 1$.

Proposition 2.13. Every division ring is a domain ($\mathcal{G}$ conversely every field is an integral domain).

Proof. Let R be a division ring & suppose $r, s \in R$ & $rs = 0$. Then assume $r \neq 0$ so r^{-1} exists, then

$s = r^{-1}rs = r^{-1}0 = 0$. Then given the immediate other case of $r = 0$ in all cases either $r = 0$ or $s = 0$.

Hence R is a domain.

Lemma 2.12. The following conditions for an integer $n \geq 2$ are equivalent:

(i) $\mathbb{Z}_n$ is an integral domain;

(ii) $\mathbb{Z}_n$ is a field;

(iii) n is prime.

Proof. (i) $\Rightarrow$ (ii): Suppose $\mathbb{Z}_n$ is a domain & let $[m]_n \in \mathbb{Z}_n$ &

$[m]_n \neq 0$. Consider the list $[0]_n, [m]_n, [1]_n, [m]_n, \dots, [n-1]_n, [m]_n$, each element in the list is distinct as, suppose otherwise

then $[i]_n[m]_n = [j]_n[m]_n$ for some distinct $i, j \in \{0, 1, \dots, n-1\}$, then $([i]_n - [j]_n)[m]_n = 0$ but $\mathbb{Z}_n$ is suppose to be a domain &

$[m]_n \neq 0$ so $[i]_n - [j]_n = [0]_n$ i.e. $[i]_n = [j]_n$, $\mathcal{G}$. Then as there are n elements of the list each in $\mathbb{Z}_n$, then

$\{[0]_n, [1]_n, [m]_n, \dots, [(n-1)]_n\} = \mathbb{Z}_n$. So some multiple of $[m]_n$ say $[r]_n \in \mathbb{Z}_n$ is s.t.

$[m]_n[r]_n = [r]_n[m]_n = [1]_n$.

(ii) $\Rightarrow$ (iii): Suppose $\mathbb{Z}_n$ is a field then suppose $m, l \in \mathbb{N}$ are s.t.

$ml = n$, so $0 < m \leq l \leq n$. Then $[m]_n[l]_n = [ml]_n = [n]_n = [0]_n$ so as fields are domains,

$[m]_n = [0]_n = [n]_n$ or $[l]_n = [0]_n = [n]_n$ & so, by the equivalence class definition,

$m = 0$ or $l = n$ as neither are 0. So if $m = n$, $l = 1$ & if $l = n$, $m = 1$, so the only factors of n are 1 and n in $\mathbb{N}$, i.e. n is prime.

(iii) $\Rightarrow$ (i): Suppose n is prime & let $[m]_n, [l]_n \in \mathbb{Z}_n$ s.t. $[m]_n[l]_n = 0$ i.e. $[ml]_n = [0]_n$ definitionally

$n | ml$ so by Th^m 1.10 $n | m$ or $n | l$ i.e. $[m]_n = [0]_n$ or $[l]_n = [0]_n$. So with $\mathbb{Z}_n$'s commutativity, $\mathbb{Z}_n$ is an integral domain. $\square$

Proposition T1. In a domain R if $ab = ac$ or $ba = ca$ for $a, b, c \in R$ $\mathcal{G}$ $a \neq 0$ then $b = c$.

Proof. If $ab = ac$, then

$a(b - c) = 0$. If $ba = ca$, then

$(b - c)a = 0$. Since $a \neq 0$ and R is a domain, $b - c = 0$.

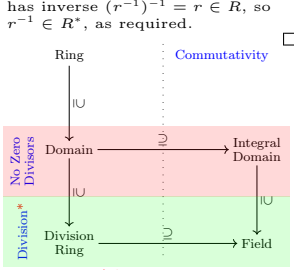
Lemma 2.15. In a ring R , the set of units R^* forms a group under multiplication.

Proof. (G0): $1 \cdot 1 = 1$ so $1 \in R^*$ i.e. $R^* \neq \emptyset$; (G1): given (R2) we have associativity & given $r, s \in R^*$,

$(rs)(s^{-1}r^{-1}) = r(s^{-1})r^{-1} = 1$ & $(s^{-1}r^{-1})(rs) = s^{-1}(r^{-1})s = 1$, so $rs \in R^*$ as required; (G2): (R4);

& (G3): Let $r \in R^*$ so r is a unit

so $r^{-1} \in R^*$ exists, which itself has inverse $(r^{-1})^{-1} = r \in R$, so $r^{-1} \in R^*$, as required. $\square$



Lemma 3.3. Suppose $\phi : R \rightarrow S$ is an isomorphism. Then: (i) $\phi(1_R) = 1_S$; (ii) $\phi(0_R) = 0_S$; (iii) $\phi(-r) = -\phi(r)$, $\forall r \in R$; (iv) $r \in R$ is invertible iff $\phi(r) \in S$ is invertible and, in that case, $(\phi(r))^{-1} = \phi(r^{-1})$; (v) $r \in R$ is nilpotent iff $\phi(r)$ is nilpotent and, in that case, they have the same index of nilpotence.

Proof. (i): Let $s \in S$. Since ϕ is bijective, $s = \phi(r)$ for some $r \in R$. Then

$\phi(1_R)s = \phi(1_R)\phi(r) = \phi(r) = s$ and similarly $s\phi(1_R) = s$, hence $\phi(1_R) = 1_S$.

(ii): $\phi(0_R) + 1_S = \phi(0_R) + \phi(1_R) = \phi(1_R) = 1_S$, so $\phi(0_R) = 0_S$.

(iii): $\phi(-r) + \phi(r) = \phi(0_R) = 0_S$, so $\phi(-r) = -\phi(r)$.

(iv): If r is invertible then $\phi(r^{-1})\phi(r) = \phi(r^{-1}r) = 1_S$ and $\phi(r)\phi(r^{-1}) = \phi(rr^{-1}) = 1_S$, so $\phi(r)$ is invertible with $(\phi(r))^{-1} = \phi(r^{-1})$. Conversely, apply this to $\phi^{-1} : S \rightarrow R$.

(v): Since $(\phi(r))^{-1} = \phi(r^{-1})$, $r^n = 0_R$ iff $(\phi(r))^n = 0_S$. Thus the least such n is the same. $\square$

Lemma 3.7. Suppose $\vartheta : R \rightarrow S$ is a homomorphism. Then: (i) $\vartheta(0) = 0$; (ii) $\vartheta(-r) = -\vartheta(r)$, $\forall r \in R$; (iii) if $r \in R$ is invertible, then $\vartheta(r) \in S$ is invertible and $(\vartheta(r))^{-1} = \vartheta(r^{-1})$; (iv) if $r \in R$ is nilpotent, then $\vartheta(r)$ is nilpotent with index $\leq$ the index of r ; (v) $\text{Im}(\vartheta)$ is a subring of S .

Proof. (i): As in Lem^a 3.3(ii). (ii): As in Lem^a 3.3(iii). (iii): If r is invertible then

$\vartheta(r^{-1})\vartheta(r) = \vartheta(r^{-1}r) = 1_S$ and similarly $\vartheta(r)\vartheta(r^{-1}) = 1_S$, so $(\vartheta(r))^{-1} = \vartheta(r^{-1})$. (iv): If r is nilpotent of index n , then

$\vartheta(r)^n = \vartheta(r^n) = \vartheta(0) = 0$, so $\vartheta(r)$ has index at most n . (v): Since $\vartheta(1_R) = 1_S, 1_S \in \text{Im}(\vartheta)$. Let $s = \vartheta(r)$, $s' = \vartheta(r')$ $\in \text{Im}(\vartheta)$. Then $s + s' = \vartheta(r + r')$, and $ss' = \vartheta(rr')$, and $s = -\vartheta(r) = \vartheta(-r)$, so by Lem^a 1.18, $\text{Im}(\vartheta)$ is a subring of S . $\square$

Lemma 3.10. (i) If $\vartheta : R \rightarrow S$ $\mathcal{G}$ $\beta : S \rightarrow T$ are homomorphisms then $\beta \circ \vartheta : R \rightarrow T$ is a homomorphism. (ii) If β, ϑ are embeddings then $\beta \circ \vartheta$ is an embedding. (iii) If β, ϑ are homomorphisms and $\beta \circ \vartheta$ is an embedding then ϑ is an embedding.

Proof. (i): Let $r, r' \in R$. Then $\beta \circ \vartheta(r + r') = \beta(\vartheta(r + r')) = \beta \circ \vartheta(r) + \beta \circ \vartheta(r')$

$= \beta \circ \vartheta(r) + \beta \circ \vartheta(r')$

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Otherwise choose non-zero $g \in I$ of least degree. Let $f \in I$. By Theorem 5.1, $f = qg + r$ with $r = 0$ or $\deg(r) < \deg(g)$. Since $r = f - qg \in I$, minimality of $\deg(g)$ forces $r = 0$. Hence every $f \in I$ is a multiple of g , so $I = (g)$. $\square$

Lemma 6.2. Let K be a field and $f \in K[X]$ be irreducible. Then $\langle f \rangle$ is a maximal ideal of $K[X]$.

Proof. Let $\langle f \rangle \leq I \leq K[X]$. By Th^m 5.8, $I = (g)$. Since $f \in (g)$, $f = qg$ for some $q \in K[X]$. As f is irreducible, q or g is invertible. If q is invertible, then $(g) = \langle f \rangle$. If g is invertible, then $(g) = K[X]$. Hence $\langle f \rangle$ is maximal. $\square$

Theorem 6.3. (Kronecker's Theorem) Let K be a field and $f \in K[X]$ be irreducible of degree n . Define $L = K[X]/\langle f \rangle$. Then: (i) L is a field and $\pi : K[X] \rightarrow L$ induces an embedding $\pi : K \rightarrow L$; (ii) $\alpha = X + \langle f \rangle$ is a root of f in L ; (iii) L is a vector space over K of dimension n , with basis $\{1, \alpha, \dots, \alpha^{n-1}\}$.

Proof. (i): By Lemma 6.2, $\langle f \rangle$ is maximal, so L is a field by Corollary 4.13. Define $\phi : K \rightarrow L$ by $\phi(a) = a + \langle f \rangle$. This is a homomorphism. If $\phi(a) = \phi(b)$, then $a - b \in \langle f \rangle$. Since $\deg(f) \geq 1$, this forces $a - b = 0$, so ϕ is injective.

(ii): Let $f = b_0 + b_1X + \dots + b_nX^n$. For $\alpha = X + \langle f \rangle$, $f(\alpha) = b_0 + b_1\alpha + \dots + b_n\alpha^n = f + \langle f \rangle = (f) = 0_L$. Thus α is a root of f in L . (iii): If $g + \langle f \rangle \in L$, divide g by f : $g = qf + r$, with $r = 0$ or $\deg(r) < n$. Writing $r = a_0 + a_1X + \dots + a_{n-1}X^{n-1}$, we get $g + \langle f \rangle = r + \langle f \rangle = a_0 + a_1\alpha + \dots + a_{n-1}\alpha^{n-1}$. Thus L is spanned by $1, \alpha, \dots, \alpha^{n-1}$. Linear independence follows since any non-zero polynomial of degree $< n$ cannot lie in $\langle f \rangle$. $\square$

Definitions

Definition 1.4. $\forall a, b \in \mathbb{Z}$, $\gcd(a, b) = \max\{n \in \mathbb{Z} : n|a \ \& \ n|b\}$, noting $\gcd(0, 0) = 0$ and if $\gcd(a, b) = 1$ then a & b are coprime.

Definition 1.6. $\forall a, b, n \in \mathbb{Z}$ with $n \geq 2$, a is **congruent to** b **modulo** n iff $n|a - b$, written $a \equiv b \pmod n$. Then define the congruence class of a w.r.t. n as: $[a]_n = \{b \in \mathbb{Z} : b \equiv a \pmod n\} = \{b \in \mathbb{Z} : n | (b - a)\} = \{a + kn : k \in \mathbb{Z}\}$. We thus have:

$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\} = \{[0]_n, [1]_n, \dots, [n-1]_n\}$. Note: $b \equiv a \pmod n$ iff $b = a + kn$ iff $[a]_n = \{a + nk : k \in \mathbb{Z}\}$ iff $[a]_n = [a + kn]_n$.

Definition 1.14. A **ring** is a set R with two binary operations $+$ & $\times$, defined on R , satisfying the following:

- (R1) $(R, +)$ is an abelian group with identity 0.
- (R2) $\times$ is associative.
- (R3) $\times$ distributes over $+$ i.e.: $a(b + c) = ab + ac$, $(a + b)c = ac + bc$, $\forall a, b, c \in R$
- (R4) The multiplicative identity $1 \in R$ & $1 \neq 0$

Example. With $n \geq 2$, $M_n(R)$ is the ring of $n \times n$ matrices, a non-commutative ring. Note: the only two-sided ideals of $M_n(F)$, F a field, are $M_n(F)$ & $\{0\}$ but this isn't a division ring hence even relaxing commutative in Pro^v 3.22 doesn't yield the same pro^v to division ring.

Example. For X a set $\mathcal{P}(X)$ with $A + B = (A \cup B) \setminus (A \cap B)$, $AB = A \cap B$ & $0 = \emptyset$, $-A = A$, $A, B \in \mathcal{P}(X)$. **Definition 1.17.** Let R be a ring & $S \subseteq R$. Then S is a **subring** of R if S is a ring w.r.t. the same addition & multiplication as R & contains the multiplicative identity of R , 1_R .

Definition 1.21. Let R be a ring. The **ring of polynomials** $R[X] = \{\sum_{i \geq 0} a_i X^i : \text{a finite sum } a_i \in R\}$ with addition & multiplication defined as:

$$\sum_{i \geq 0} a_i X^i + \sum_{i \geq 0} b_i X^i = \sum_{i \geq 0} (a_i + b_i) X^i$$
$$\sum_{i \geq 0} a_i X^i \times \sum_{j \geq 0} b_j X^j = \sum_{i \geq 0} (\sum_{j=0}^i a_j b_{i-j}) X^i$$

(taking missing coefficients as 0) $0 = \sum_{i \geq 0} 0X^i$ & $1 = 1X^0 + \sum_{i \geq 1} 0X^i$. Then the **degree** of a polynomial is the greatest $i \in \mathbb{N} \cup \{0\}$ s.t. $a_i \neq 0$ or if no such i exist $\deg(0) = -\infty$. More generally, we have: $R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$ $\forall f, g \in R[X]$, $\deg(f + g) \leq \max(\deg(f), \deg(g))$ & $\deg(fg) \leq \deg(f) + \deg(g)$. **Definition 1.24.** If R_1 & R_2 are rings then the **Cartesian product** $R_1 \times R_2$ with

the $+$ & $\times$ defined element wise is a ring.

Definition 2.1. For $a \in R$ define: for $n \in \mathbb{Z}^+$, $n \cdot a = \underbrace{a + a + \dots + a}_n$ for $n = 0$,

$n \cdot a = 0 \cdot a = 0$; & for $n \in \mathbb{Z}^-$, $n \cdot a = (-n) \cdot (-a)$.

The **Characteristic**, $\text{char}(R)$, of a ring R is the least positive integer n s.t. $n \cdot 1 = 0$, or if no exist $\text{char}(R) = 0$.

Note: $(n + m) \cdot a = n \cdot a + m \cdot a$, $n \cdot (a + b) = n \cdot a + n \cdot b$, & $n \cdot (m \cdot a) = (nm) \cdot a$, all easily verifiable.

Definition 2.3. A non-zero element $r \in R$ is a **zero-divisor** if $\exists s \in R$ a non-zero element s.t. $rs = 0$ or $sr = 0$.

A ring R is a domain if, $\forall r, s \in R$: $rs = 0 \implies r = 0$ or $s = 0$ i.e. a ring with no zero divisors is a domain & vice versa.

Definition 2.9. A **division ring** is a ring where every non-zero element has a right & left inverse, i.e.: $\forall r \in R \setminus \{0\}, \exists s, t \in R$ s.t. $rs = 1 = tr$ with s, t being the right & left inverses, respectively.

Note: this immediately collapse, due to Le^m 2.11, to just there exists an inverse as $s = t$.

An **invertible** element r is called a **unit**.

Example. Division rings are not all fields e.g. the quaternions $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ where $i^2 = j^2 = k^2 = -1$ & $ij = k, jk = i, \& \ ki = j$, this is non-commutative as $-ij = -k$ vs $jik = ji$ so $ij \neq ji$ and this is a division ring as for $z = a + bi + cj + dk$ $z^{-1} = \frac{a-bi-cj-dk}{a^2+b^2+c^2+d^2}$, taking $z \neq 0$.

Definition 2.16. An element r of a ring R is **nilpotent** if $\exists n \in \mathbb{Z}^+$ s.t. $r^n = 0$, and the least such n is the **index of nilpotence** of r .

An element $r \in R$ is **idempotent** if $r^2 = r$.

Note: In any ring R , $0R$ is nilpotent with index 1 and $0R$ & $1R$ are idempotent.

Definition 3.1. If R & S are rings then an **isomorphism** from R to S is a bijection $\phi : R \rightarrow S$ s.t. $\forall r, r' \in R$:

$$\phi(r + r') = \phi(r) + \phi(r'),$$
$$\phi(rr') = \phi(r)\phi(r')$$

$\exists \phi : R \rightarrow S$ an isomorphism iff R & S are **isomorphic**, written $R \simeq S$.

Example. For 4 element rings $\mathbb{Z}_4 \not\simeq \mathbb{Z}_2 \times \mathbb{Z}_2$, since given $[2]_4 \neq 0$ and $[2]_4^2 = [0]_4$, $[2]_4$ is nilpotent, where as there are no non-zero nilpotent elements in $\mathbb{Z}_2 \times \mathbb{Z}_2$, hence the result by Le^m 3.3.

Definition 3.6. If R & S are rings then a **homomorphism** from R to S is a map $\vartheta : R \rightarrow S$ s.t. $\forall r, r' \in R$:

$$\vartheta(r + r') = \vartheta(r) + \vartheta(r'),$$
$$\vartheta(rr') = \vartheta(r)\vartheta(r'),$$
$$\vartheta(1_R) = 1_S.$$

Example. The map $\mathbb{Z} \rightarrow \mathbb{Z}_n$ by $m \mapsto [m]_n$ is a homomorphism. Also, $\vartheta : \mathbb{Z}_{kn} \rightarrow \mathbb{Z}_n$ by $\vartheta([m]_{kn}) = [m]_n$ is a homomorphism, well definedness must be shown, and is true as; let $[m]_{kn} = [m']_{kn} \in \mathbb{Z}_{kn}$ then $kn | m - m'$ so $n | m - m'$ thus $[m]_n = [m']_n$.

Definition 3.9. An **embedding** or **monomorphism** is an injective homomorphism. There is a trivial embedding along each arrow of $\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R} \hookrightarrow \mathbb{C} \hookrightarrow \mathbb{C}[X]$.

Example. With R a ring, the map $R[X] \rightarrow R[X]$ by $p(X) \mapsto p(q(X))$ for some $q(X) \in R[X]$.

Definition 3.12. If $\vartheta : R \rightarrow S$ is a ring homomorphism, then the **kernel** of ϑ is defined as $\ker(\vartheta) = \{r \in R : \vartheta(r) = 0_S\}$.

Definition 3.16. An **automorphism** of a ring is an isomorphism from R to itself.

Example. $\mathbb{Z}[i] \rightarrow \mathbb{Z}[i]$ by $a + bi \mapsto a - bi$ is an automorphism.

Definition 3.18. An **ideal** of a ring R is a subset $I \subseteq R$ s.t.: $0 \in I$; $a + b \in I, \forall a, b \in I$; & $ar, ra \in I, \forall a \in I, \forall r \in R$. We write $I \triangleleft R$ to mean I is an ideal of R .

Definition 3.19. If $a \in R$ then $\{r_1as_1 + \dots + r_nas_n : n \geq 1, r_i, s_i \in R\}$ is an ideal containing a and is the smallest ideal of R containing a . It is called the **principal ideal generated by** a , denoted $\langle a \rangle$. If R is commutative then $\langle a \rangle = \{ar : r \in R\}$. A **principal ideal** is one which can be generated by a single element.

Examples 3.20. (i) In every ring $\langle 0 \rangle = \{0\}$ is the smallest ideal, the **trivial ideal**. (ii) In every ring $\langle 1 \rangle = R$ is an ideal and every other ideal $\neq R$ is a **proper ideal**. (iii) If $n \in \mathbb{Z}$ then

$\langle n \rangle = \{nk : k \in \mathbb{Z}\}$. (iv) If $p \in R[X]$ then $\langle p \rangle$ is the set of polynomials with p as a factor.

Definition 3.21. A **right ideal** is defined as for a general ideal but with the third condition replaced by the weaker condition: $a \in I$ and $r \in R$ implies $ar \in I$. A **left ideal** is defined similarly. If $a \in R$, the **principal right ideal** generated by a is $\{ar : r \in R\}$, denoted aR .

Example. Let $R = M_2(\mathbb{Z})$ and $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ then by simple multiplication the principle left and right ideals are $\{\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} : a, b \in \mathbb{Z}\}$ and $\{\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{Z}\}$ whilst $\langle A \rangle = M_2(\mathbb{Z})$ as $I_2 = A + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \langle A \rangle$

Note: If $1 \in I$ with $I \triangleleft R$ the $\forall r \in R : r = r1 \in I$, so $I = R$.

Definition 3.22. Let $A \subseteq R$ and R be a commutative ring, then $\{a_1r_1 + \dots + a_nr_n : n \geq 1, a_i \in A, r_i \in R\}$ is the **ideal generated by** A .

Example. Working in $\mathbb{Z}[X]$ the ideal $I = \langle 2, X \rangle := \langle \{2, X\} \rangle$ is not a principal ideal. Suppose $I = \langle p \rangle$. Since $2 \in I$, $2 = pq$ for some $q \in \mathbb{Z}[X]$. As $\mathbb{Z}$ is a domain, $0 = \deg(2) = \deg(pq) = \deg(p) + \deg(q)$, so $p, q \in \mathbb{Z}$ and $p = \pm 1$ or $p = \pm 2$. If $p = \pm 1$, then $1 = 2r_1 + Xr_2$ for some $r_1, r_2 \in \mathbb{Z}[X]$, impossible since the RHS has even constant term. Hence $p = \pm 2$. But $X \in I$, so $X = 2r_3$ for some $r_3 \in \mathbb{Z}[X]$, impossible since $2r_3$ has even coefficients. Thus I is not principal.

Definition 3.26. We can also define a **product of ideals**: if $I, J \triangleleft R$, then their product is the ideal generated by $\{ab : a \in I, b \in J\}$. This set will not in general be closed under addition, hence we say "generated by". Powers of an ideal are defined inductively by $I^{n+1} = I^n I$.

Definition 4.1. Let R be a ring and let I be a proper ideal. Let R/I denote the set of cosets of I in the additive group $(R, +)$:

$$R/I = \{r + I : r \in R\}.$$

Define operations $+$ & $\times$ on R/I by $(r + I) + (s + I) = (r + s) + I$, $(r + I)(s + I) = rs + I$.

Definition 4.1. The ring R/I is called the **factor ring** (or **quotient ring**) of R by I .

Example. $\mathbb{Z}/\langle n \rangle = \mathbb{Z}_n$. **Exercise.** Write out the elements of the factor ring $\mathbb{Z}_6/\langle [3]_6 \rangle$. What is this factor ring isomorphic to?

Definition 4.4. The map $\pi : R \rightarrow R/I$, $\pi(r) = r + I$, is called the **canonical projection**, or **canonical surjection**.

Example 4.6. Take $R = \mathbb{Z}$ and $I = \langle 6 \rangle = 6\mathbb{Z}$, so the canonical projection is $\pi : \mathbb{Z} \rightarrow \mathbb{Z}_6$. Let $\vartheta : \mathbb{Z} \rightarrow \mathbb{Z}/\langle 2 \rangle = \mathbb{Z}_2$ be the canonical projection. Then $\ker(\vartheta) = \langle 2 \rangle \supseteq \langle 6 \rangle$, so by Le^m 4.4 there is a unique homomorphism ϑ' with $\vartheta' \circ \pi = \vartheta$. This is the map $\vartheta' : \mathbb{Z}_6 \rightarrow \mathbb{Z}_2$ already seen in Examples 3.8.

Example 4.7. Let $R = \mathbb{R}[X]$ and $I = \langle X^2 + 1 \rangle$. Let $\pi : \mathbb{R}[X] \rightarrow \mathbb{R}[X]/\langle X^2 + 1 \rangle$ be the canonical surjection. Let $\vartheta : \mathbb{R}[X] \rightarrow \mathbb{C}$ be evaluation at i , $\vartheta(p(X)) = p(i)$. Then $p \in \ker(\vartheta)$ iff $X^2 + 1 \mid p$, so $\ker(\vartheta) = \langle X^2 + 1 \rangle$. By Le^m 4.4, there is a unique map $\vartheta' : \mathbb{R}[X]/\langle X^2 + 1 \rangle \rightarrow \mathbb{C}$ through which ϑ factors. It is injective and surjective, hence an isomorphism. Thus $\mathbb{R}[X]/\langle X^2 + 1 \rangle \simeq \mathbb{C}$.

Notation: if $I \subseteq J$ are ideals then usually write $I \leq J$ (or $J \geq I$) instead of subset notation, with the same meaning. **Example 4.9.** The ideals of $\mathbb{Z}_6 = \mathbb{Z}/\langle 6 \rangle$ are in one-to-one correspondence with the ideals of $\mathbb{Z}$ containing $\langle 6 \rangle$. There are 4 of each: $\langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 6 \rangle$ correspond to $\pi(1) = \mathbb{Z}_6 = \{[0]_6, [1]_6, [2]_6, [3]_6, [4]_6, [5]_6\}$, $\pi(2) = \langle [2]_6 \rangle = \{[0]_6, [2]_6, [4]_6\}$, $\pi(3) = \langle [3]_6 \rangle = \{[0]_6, [3]_6\}$, $\pi(6) = \langle [0]_6 \rangle = \{[0]_6\}$.

The notation J/I is also used instead of πJ for the image of J in R/I .

Example 4.11. Since $\langle 6 \rangle \leq \langle 2 \rangle$, Th^m 4.10 gives $(\mathbb{Z}/\langle 6 \rangle)/(\langle 2 \rangle/\langle 6 \rangle) \simeq \mathbb{Z}/\langle 2 \rangle$.

Definition 4.12. An ideal I of a ring R is **maximal** if it is proper and for any ideal J with $I \leq J \leq R$, either $J = I$ or $J = R$.

Definition 4.14. A proper ideal I of a commutative ring R is **prime** if whenever $r, s \in R$ and $rs \in I$, then $r \in I$ or $s \in I$.

Example 4.15. All ideals of $\mathbb{Z}$ are principal. The prime ideals are precisely the ideals generated by a prime number or zero.

Example 5.2. In $\mathbb{Z}_5[X]$, $X^4 + 2X^3 + X + 4 = (X^2 + 2X + 3)(X^2 + 3X + 3) + X$.

Definition 5.3. An element $a \in K$ is a **root** (or **zero**) of $f \in K[X]$ iff $f(a) = 0$.

Definition 5.5. The **greatest common divisor** of polynomials f, g is a polynomial d such that $d \mid f$ and $d \mid g$, and if $h \mid f$ and $h \mid g$, then $h \mid d$. This determines d up to a non-zero scalar; for uniqueness take d **monic**.

Division Algorithm for Polynomials. For $f, g \in K[X]$, $g \neq 0$, repeatedly apply Th^m 5.1: $f = q_1g + r_1$, $\deg(r_1) < \deg(g)$, $g = q_2r_1 + r_2$, $\deg(r_2) < \deg(r_1)$, $r_1 = q_3r_2 + r_3$, $\deg(r_3) < \deg(r_2)$, $\vdots$

$r_{n-1} = q_{n+1}r_n$. Since degrees strictly decrease, some $r_{n+1} = 0$. Then $\gcd(f, g) = \gcd(g, r_1) = \dots = \gcd(r_{n-1}, r_n) = r_n$ up to multiplication by a non-zero scalar; take the monic version.

Tracking back expresses $\gcd(f, g) = kf + lg$ for some $k, l \in K[X]$. **Factorisation.** Here R is commutative. **Definition 5.9.** (i) $r \in R$ is **irreducible** if r is not invertible and whenever $r = st$, either s or t is invertible. (ii) $r, s \in R$ are **associated** if $s = ur$ for some invertible $u \in R$.

Examples 5.10. (i) In $\mathbb{Z}$, irreducibles are $\pm p$ for p prime; r, s are associated iff $r = \pm s$. (ii) In $\mathbb{R}[X]$, irreducibles are non-constant polynomials with no factor g satisfying $1 \leq \deg(g) < \deg(f)$.

Definition 5.11. A commutative domain R is a **UFD** if every non-zero, non-invertible element factors uniquely into irreducibles, up to rearrangement and associated factors. **Example 5.12.** $\mathbb{Z}$ is a UFD. For example, $-24 = (-2) \cdot 2 \cdot (-2) \cdot (-3)$, unique up to signs and ordering.

Example 5.13. $\mathbb{Z}[\sqrt{-5}]$ is not a UFD since $6 = 3 \cdot 2 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, with irreducible factors not matching up to associates. **Definition 5.14.** A **Principal Ideal Domain (PID)** is a commutative domain in which every ideal is principal.

Examples 5.15. $\mathbb{Z}$ is a PID; if K is a field, $K[X]$ is a PID by Th^m 5.8. Every PID is a UFD. If R is a UFD then so is $\mathbb{R}[X]$, hence $\mathbb{R}[X_1, \dots, X_n]$.

Example. $\mathbb{Z}[X]$ is a UFD but not a PID: it is a UFD since $\mathbb{Z}$ is, but $\langle 2, X \rangle$ is not principal.

Examples 5.16. (i) $X^3 + X^2 + 1$ is irreducible in $\mathbb{Z}_3[X]$: a reducible cubic has a linear factor, but 0, 1 are not roots. (ii) $X^4 + 2X^3 + X + 1$ is irreducible in $\mathbb{Z}_3[X]$: no roots, and comparing coefficients in two possible quadratics gives no factorisation. **Example 6.1.** Take $p = X^2 - 2 \in \mathbb{Q}[X]$. An extension field containing a root is $\mathbb{R}$, or more efficiently $\mathbb{Q}[\sqrt{2}]$.

To produce a root for a polynomial p , it is enough to produce a root of one irreducible factor of p . Hence focus on irreducible polynomials.

Example 6.4. (i) If $K = \mathbb{Q}$ and $f = X^2 - 2$, then $\mathbb{Q}[X]/\langle X^2 - 2 \rangle \simeq \mathbb{Q}[\sqrt{2}]$. (ii) If $K = \mathbb{R}$ and $f = X^2 + 1$, every element of $\mathbb{R}[X]/\langle X^2 + 1 \rangle$ has form $a_0 + a_1i$, using i for $X + \langle f \rangle$, so $\mathbb{R}[X]/\langle X^2 + 1 \rangle \simeq \mathbb{C}$.

Example 6.5. If $f = X(X - 1)$, then $K[X]/\langle X(X - 1) \rangle$ is not a field. With $\alpha = X + \langle f \rangle$, neither α nor $\alpha - 1$ is zero, but $\alpha(\alpha - 1) = 0$. So the quotient is not even a domain.

Example 6.6. For $X^3 + X + 1 \in \mathbb{Z}_2[X]$, there are no roots in $\mathbb{Z}_2$, so it is irreducible. Let $\alpha = X + \langle f \rangle$ and $L = \mathbb{Z}_2[X]/\langle f \rangle$. By Kronecker, L is a field with 2^3 elements:

$\{0, 1, \alpha, \alpha^2, 1 + \alpha, 1 + \alpha^2, \alpha + \alpha^2, 1 + \alpha + \alpha^2\}$. In L , $\alpha^3 + \alpha + 1 = 0$, so $\alpha^3 = -1 = 1$, which determines the addition and multiplication tables.

